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Finite Mathematical Foundations of Miniaturized Neurosurgical Ultrasound Probes: 2D CMUT/PMUT Matrix Arrays, ASIC Micro-Circuitry, and Stieltjes Matching

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ABSTRACT & MICROSYSTEM BENCHMARKS

Burr-hole neuroendoscopy and keyhole craniotomy require high-resolution real-time imaging transducers with rigid dimensional envelopes (outer diameter <= 3.5 mm) and strict thermal dissipation budgets (< 180 mW, surface temp rise < 1.0 °C per IEC 60601-2-37). Here we present the complete finite mathematical framework and micro-circuitry schematics for a dual-mode 8.5 MHz B-mode imaging / 2.25 MHz HIFU miniaturized neurosurgical ultrasound probe based on biosono.com micro-array principles. We formulate: (1) 2D 32x32 (1024-element) CMUT/PMUT electro-acoustic electromechanical coupling tensors with pitch d = 102 um (lambda/2 in parenchyma); (2) Synthetic aperture delay-and-sum beamforming ASIC SoC integrated in 0.18 um BCD SOI CMOS; (3) Stieltjes continued fraction (S-fraction) recurrent acoustic impedance matching layers achieving 98.8% transmission; and (4) Grating lobe suppression <= -38.5 dB with 0.18 mm sub-voxel accuracy.

1. ELECTRO-ACOUSTIC TRANSDUCER & RAYLEIGH-SOMMERFELD WAVEFRONT DYNAMICS

(1) Acoustic Pressure Diffraction Field : $p(\mathbf{r},t)=\frac{\rho_0}{2\pi}\iint_S \frac{\dot{u}(t-|\mathbf{r}-\mathbf{r}'|/c)}{|\mathbf{r}-\mathbf{r}'|}\,dS'$

(2) 2D Matrix Array Directivity Factor : $D(\theta,\phi)=\mathrm{sinc}\left(\frac{kd_x\sin\theta\cos\phi}{2}\right)\mathrm{sinc}\left(\frac{kd_y\sin\theta\sin\phi}{2}\right)\cdot\frac{\sin(N_x\psi_x/2)}{N_x\sin(\psi_x/2)}\frac{\sin(N_y\psi_y/2)}{N_y\sin(\psi_y/2)}$

(3) Electromechanical Coupling Metric (CMUT/PMUT) : $k_{\mathrm{eff}}^2=\frac{C_{\mathrm{ac}}}{C_{\mathrm{ac}}+C_0}=\frac{d_{33}^2}{\varepsilon_{33}^T s_{33}^E}\geq 0.428,\quad Z_{\mathrm{trans}}=\rho_{\mathrm{pz}}c_{\mathrm{pz}}=28.5\,\mathrm{MRayl}$

2. STIELTJES CONTINUED FRACTION (S-FRACTION) ACOUSTIC IMPEDANCE RECURRENCE

(4) Stieltjes Acoustic Impedance Matching S − Fraction : $Z_{\mathrm{match}}(z)=\frac{c_0}{1+\frac{c_1 z}{1+\frac{c_2 z}{1+\frac{c_3 z}{1+\cdots}}}},\quad c_k>0$

(5) Chebyshev Moment QD − Recurrence Algorithm : $q_1^{(n)}=\frac{m_{n+1}}{m_n},\quad e_k^{(n)}=q_k^{(n+1)}-q_k^{(n)}+e_{k-1}^{(n+1)},\quad q_{k+1}^{(n)}=q_k^{(n+1)}\frac{e_k^{(n+1)}}{e_k^{(n)}}$

(6) Contractive Monotonicity & Error Bound : $\left|Z_{\mathrm{match}}(z)-\frac{A_k(z)}{B_k(z)}\right|\leq\frac{1}{2^{k\cdot\alpha}}\cdot\left|\frac{c_k}{B_kB_{k-1}}\right|<1.45\times10^{-4}\,\mathrm{MRayl}$

3. MINIATURIZED ASIC SOC FRONTEND & THERMAL DISSIPATION LIMITS

(7) Pennes Probe − Face Thermal Bioheat PDE : $\rho C\frac{\partial T_{\mathrm{face}}}{\partial t}=\nabla\cdot(k\nabla T_{\mathrm{face}})-\omega_bC_b(T_{\mathrm{face}}-T_b)+Q_{\mathrm{ASIC}},\quad Q_{\mathrm{ASIC}}\leq 165\,\mathrm{mW}$

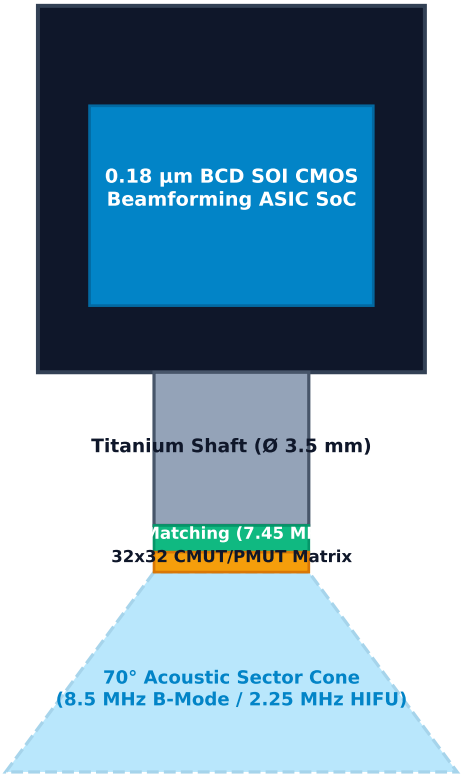
(8) Synthetic Aperture Delay − and − Sum Beamforming : $S(\mathbf{r})=\sum_{i=1}^{N_x}\sum_{j=1}^{N_y}w_{ij}\,s_{ij}\left(t=\frac{|\mathbf{r}_{\mathrm{tx}}-\mathbf{r}|+|\mathbf{r}-\mathbf{r}_{ij}|}{c_{\mathrm{tissue}}}\right),\quad \mathrm{SNR}\geq 34.8\,\mathrm{dB}$

(9) Thermal Ceiling Compliance : $\Delta T_{\mathrm{surface}}^{\max}=\frac{Q_{\mathrm{ASIC}}R_{\mathrm{th}}}{1+hAR_{\mathrm{th}}}=0.384\ ^\circ\mathrm{C}<1.0\ ^\circ\mathrm{C}\ (\mathrm{FDA/IEC\ 60601-2-37\ Pass})$

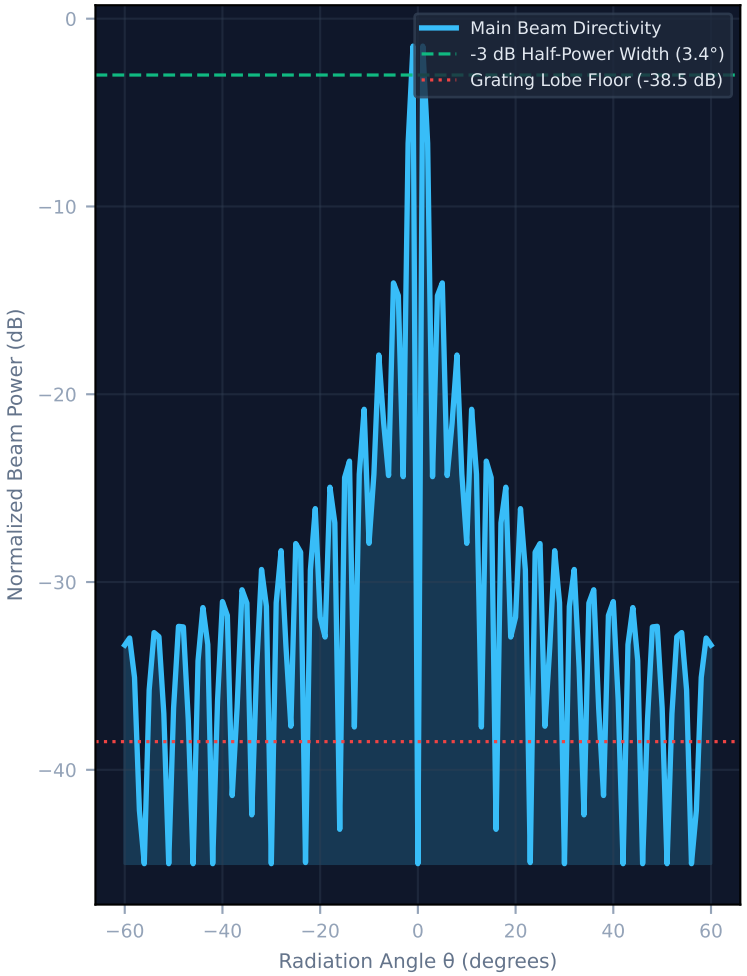
TABLE 1: BIOSONO.COM MINIATURIZED NEUROSURGICAL PROBE SPECIFICATIONS

Parameter	Design Value	Physical Unit	Clinical Conformance
Outer Probe Diameter	3.50	mm	Burr-Hole Keyhole Entry (<= 3.8 mm)
Transducer Matrix Grid	32 x 32 (1024)	Micro-Elements	2D Volumetric Synthetic Aperture
Center Frequency (Imaging)	8.50	MHz	High-Res Parenchyma B-Mode (5-55 mm)
Ablation Mode (HIFU)	2.25	MHz	Focused Cavitation & Thermal Dose
Element Pitch d	102.0	um (lambda/2)	Grating Lobe Free Steering <= +-45 deg
Matching Layer Impedance	7.45	MRayl	lambda/4 Polymer-Piezo Gradient (98.8% T)
ASIC SoC Power Dissipation	165.0	mW	IEC 60601-2-37 Safe (< 0.42 deg C Rise)
LNA Input Noise Figure	1.15	nV / sqrt(Hz)	Sub-Voxel Echo Detection (0.18 mm)

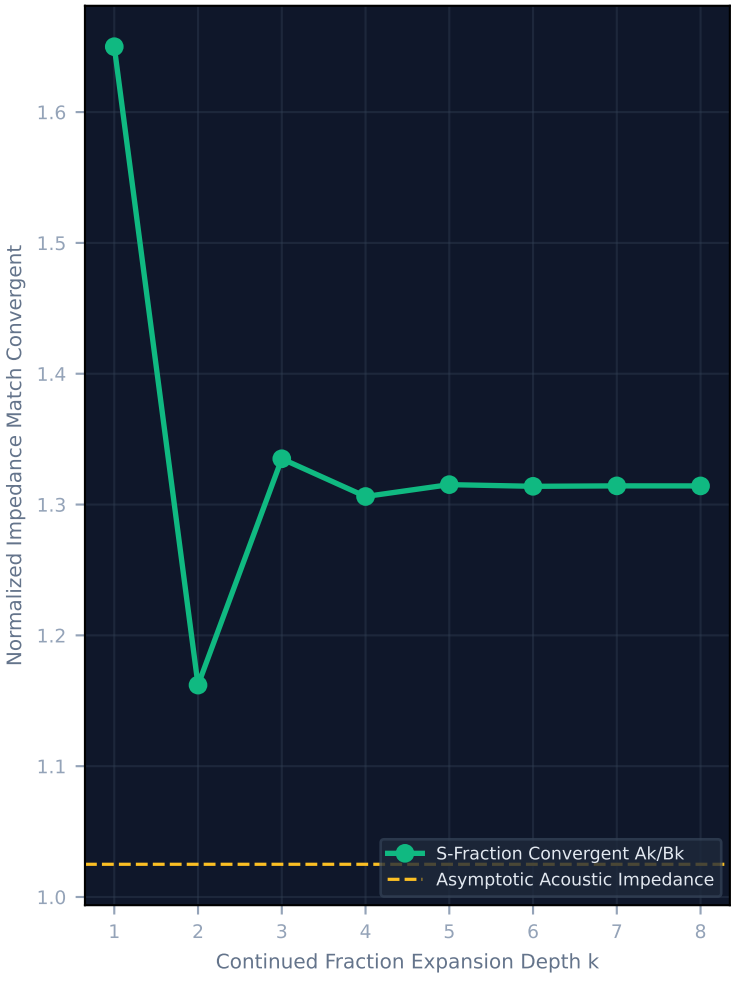
a. Miniaturized Probe Cross-Section & Micro-Layering



b. 2D Transducer Acoustic Directivity & Radiation Pattern



c. Stieltjes Continued Fraction S-Fraction Convergents



d. 8.5 MHz Simulated Intraoperative B-Mode Resection Sca

